

Longest Uncommon Subsequence

Authored by: Arpit Raj, LNMIIT Jaipur

Copyright © July 25, 2026. Authored by Arpit Raj, LNMIIT Jaipur.

Problem Understanding

Suppose `strs = ["aba", "cdc", "eae"]`

We need to find the longest string that is a subsequence of exactly one string.

Suppose `"ab"` is uncommon.

Then `"aba"` is also uncommon because `"ab"` is a subsequence of `"aba"`. Since we're looking for the longest uncommon subsequence, we'd always prefer the whole string. So we only need to check each original string.

Examples

Example 1: `["aba", "cdc", "eae"]`

Take "aba". Ask:

Is "aba" a subsequence of "cdc"? **No.**

Is "aba" a subsequence of "eae"? **No.**

So "aba" appears as a subsequence in only one string (itself).

Length = 3.

Example 2: ["aaa", "aaa", "aa"]

Check "aaa" (first one). Is it a subsequence of the second "aaa"? **Yes**. So reject it.

Check second "aaa". Is it a subsequence of the first? **Yes**. Reject.

Check "aa". Is "aa" a subsequence of "aaa"? **Yes**. Reject.

No valid string remains.

Answer = -1.

Is 'a' a subsequence of 'b'?

Use two pointers.

```
a : a b c
    ^
b : a x b y c z
    ^
```

Both characters are 'a'. Move both pointers.

```
a : a b c
    ^
b : a x b y c z
    ^
```

b[1]='x' doesn't match 'b'. Move only b.

```
a : a b c
    ^
b : a x b y c z
      ^
```

Now 'b' matches. Move both.

```
a : a b c
    ^
b : a x b y c z
      ^
```

'y' doesn't match. Move b.

```
a : a b c
    ^
b : a x b y c z
      ^
```

'c' matches. Move both.

Now we've matched every character of a. **Return true.**

Implementation

```
class Solution { public: bool issub(string a, string b) { int n = a.size(); int m = b.size(); int i = 0; int j = 0; while (i < n && j < m) { if (a[i] == b[j]) { i++; j++; } else { j++; } } if (i == n) return true; // agar match krte krte saare match ho gaye to a is a sub of b else return false; } int findLUSlength(vector<string>& strs) { int n = strs.size(); int ans = -1; // try every substring as the answer for (int i = 0; i < n; i++) { bool flag = true; for (int j = 0; j < n; j++) { if (i == j) continue; if (issub(strs[i], strs[j])) { flag = false; break; // ek bhi match mil gya matlab unique nhi ho skta , dusra try karo } } if (flag) { ans = max(ans, (int)strs[i].size()); } } return ans; } };
```